

IV. (a) Define the following:

- (i) Gross head
- (ii) Net head
- (iii) Hydraulic Efficiency
- (iv) Overall efficiency.

- (b) Pelton wheel for a hydro-electric power plant has to be designed for the following data: power to be developed is 100 MW. Net head available is 450 m. Speed = 400 rpm, jet ratio = 10, Overall efficiency = 90 %, co-efficient of velocity = 0.98 and speed ratio = 0.45. Find total quantity of water required, diameter of the jet, number of jets (nozzles) required and diameter of the wheel.

8 L4

OR

- V. (a) Define the unit quantities employed in the performance study of hydraulic turbines. Why are they important?

4 L4

- (b) The runner of a Kaplan turbine is designed to develop 9.1 MW power while operating under a net head of 5.6 m. The speed ratio based on outer diameter and flow ratio are to be 2.09 and 0.68 respectively and the outer to hub diameter ratio is 3. Assuming an overall efficiency of 86%. Find the diameter of the runner. Also find the speed and specific speed of the machine.

8 L4

- VI. (a) Draw and discuss operating characteristics of a centrifugal pump.

4 L3

- (b) The outer diameter of an impeller of a centrifugal pump is 400 mm and outlet width is 50 mm. The pump is running at 800 r.p.m. and is working against a total head of 15 m. The vanes angle at the outlet is 40° and manometric efficiency is 75%. Determine

8 L4

- (i) velocity of flow at outlet
- (ii) velocity of water leaving the vane
- (iii) angle made by the absolute velocity at outlet with the direction of motion at outlet
- (iv) discharge.

OR

- VII. (a) Define slip, percentage slip and negative slip of a reciprocating pump.

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- (b) The cylinder bore diameter of a single-acting reciprocating pump is 150 mm and its stroke is 300 mm. The pump runs at 50 r.p.m. and lifts water through a height of 25 m. The delivery pipe is 22 m long and 100 mm in diameter. Find the theoretical discharge and the theoretical power required to run the pump. If the actual discharge is 4.2 litres/sec, find the percentage slip. Also determine the acceleration head at the beginning and middle of the delivery stroke.

8

- VIII. Explain with the help of neat sketch, the principle and working of the following hydraulic devices:

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- (i) Hydraulic accumulator
- (ii) Hydraulic crane

OR

- IX. (a) Explain with the help of a neat sketch the principle and working of hydraulic coupling.

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- (b) The pressure intensity of water supplied to an intensifier is 20 N/cm^2 while the pressure intensity of water leaving the intensifier is 100 N/cm^2 . The external diameter of the sliding cylinder is 20 cm. Find the diameter of the fixed ram of the intensifier.

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Bloom's Taxonomy Levels

- 10%, L2 - 12.5%, L3 - 26.6%, L4 - 50%

$$Q = AV$$

$$L = 27$$

B.Tech. Degree IV Semester Regular/Supplementary Examination June 2022

ME 19-205-0405 HYDRAULIC MACHINERY (2019 Scheme)

Time: 3 Hours

Maximum Marks

Course Outcome

On successful completion of the course, the students will be able to:

- CO1: Understand dimensional analysis and principles of similitude.
 CO2: Apply the momentum principles to impinging jets and analyse its performance.
 CO3: Learn the working of various hydraulic turbines and their performance characteristics.
 CO4: Understand the working of roto-dynamic pumps and positive displacement pumps.
 CO5: Study the principle of working of hydraulic devices.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze,

L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A

(Answer ALL questions)

- | I. | (8 × 3 = 24) | Marks | BL |
|--|--------------|-------|----|
| (a) What is meant by geometric, kinematic and dynamic similarities? | 3 | L2 | |
| (b) Obtain an expression for the force exerted by a jet of water on the inclined plate moving in the direction of the jet. | 3 | L2 | |
| (c) What is meant by specific speed of a turbine? What is its significance? | 3 | L1 | |
| (d) State the purpose of draft tube in a reaction water turbine. Why does a Pelton wheel not possess any draft tube? | 3 | L2 | |
| (e) What is priming and explain the necessity of priming? | 3 | L1 | |
| (f) Explain ideal indicator diagram for reciprocating pumps. | 3 | L2 | |
| (g) Obtain the expression for Rankine's efficiency of a hydraulic ram. | 3 | L1 | |
| (h) What is a hydraulic intensifier? Explain its principle and working. | 3 | L2 | |

PART B

(4 × 12 = 48)

- II. (a) Show that the efficiency of a free jet striking normally on a series of flat plates mounted on a periphery of a wheel can never exceed 50%.
 (b) A jet of water having a velocity of 20 m/s strikes a curved vane, which is moving with a velocity of 10 m/s. The jet makes an angle of 20° with the direction of motion of vane at inlet and leaves at an angle of 130° to the direction of motion of vane at outlet. Calculate:
 (i) Vane angles, so that the water enters and leaves the vane without shock.
 (ii) Work done per unit weight of water striking the vane per second.

OR

- III. (a) What do you mean by dimensionless numbers? Define and explain any two of them.
 (b) Derive on the basis of dimensional analysis, suitable parameters to present the thrust developed by a propeller. Assume that the thrust P depends upon the angular velocity ω , speed of advance V, diameter D, dynamic viscosity μ , mass density ρ , elasticity of the fluid medium which can be denoted by the speed of sound in the medium C.